

```
[shobhit@localhost shobhit]$ ls -l
total 4
..... 1 shobhit shobhit 25 Mar  3 22:01 f_shobhit
[shobhit@localhost shobhit]$ cat f_shobhit
cat: f_shobhit: Permission denied
```

Figure 2: Effect of no permissions on files

- Next, add another user *test* as a member of the group *shobhit*:

```
useradd -G shobhit test
```

Add *read* permission to the owner group for the file:

```
chmod g+r f_shobhit
```

Login with the user *test* and read the file:

```
su - test
cat f_shobhit
```

The contents of the file will be displayed.

- Add another user *test1* who is neither the user owner nor a member of the group *shobhit*:

```
useradd test1
```

(See Figure 3.)

Add *read* permission to other users also:

```
chmod o+r f_shobhit
```

Login with user *test1* and try to read the file—the contents of the file will be displayed.

Similarly, *write* and *execute* permissions will work for user owner, owner group member users and other users.



Note 1: To implement *execute* permissions, *read* permissions must be provided because file contents are read and then sent for execution.

Note 2: It doesn't make sense to give *write* permissions on a file without the *read* permissions, in general, since this will simply overwrite an existing file.

Note 3: It is strongly recommended not to give *write* permissions to other users, including every unauthorised person who can use the system and, with loose permissions management, can harm your files. It is better to appoint all the authorised persons as group members and then give them *write* privileges. Advanced permissions may be implemented using ACLs and SELinux, which are out of the scope of this article.

```
[root@localhost ~]# cat /etc/group | tail -3
shobhit:x:500:test
test:x:501:
test1:x:502:
[root@localhost ~]#
```

Figure 3: Details of users just created

Effects of permissions on directories

Let us create a directory named *d_shobhit* under the */home/* Common path and clear all permissions from that directory:

```
mkdir d_shobhit
chmod a= d_shobhit
```

Now, executing the following statements from any user will show the *Permission Denied* error.

```
cd d_shobhit
ls d_shobhit
touch d_shobhit/file1
```

- Giving *read* permission to all users will allow the listing of the directory's contents but the other two operations still won't work:

```
chmod a+r d_shobhit
ls -ld d_shobhit
```

- Adding *execute* permission will allow opening of the directory and viewing its contents, but new files still can't be created:

```
chmod a+x d_shobhit
cd d_shobhit
```

- Adding *write* permission will allow the creation of new contents:

```
chmod a+w d_shobhit
touch d_shobhit/file1
```

- Now, removing *read* permission will allow the creation of a file and opening of the directory, but listing directory contents using the *ls* command will not be permitted:

```
chmod a-r d_shobhit
touch d_shobhit/file2
cd d_shobhit
```

- Keeping *read* and *write* permissions, but removing *execute* permissions will not allow the opening of the directory or the creation of contents inside it. Listing of contents will be incomplete, and a warning will be issued